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What is Participatory GIS?

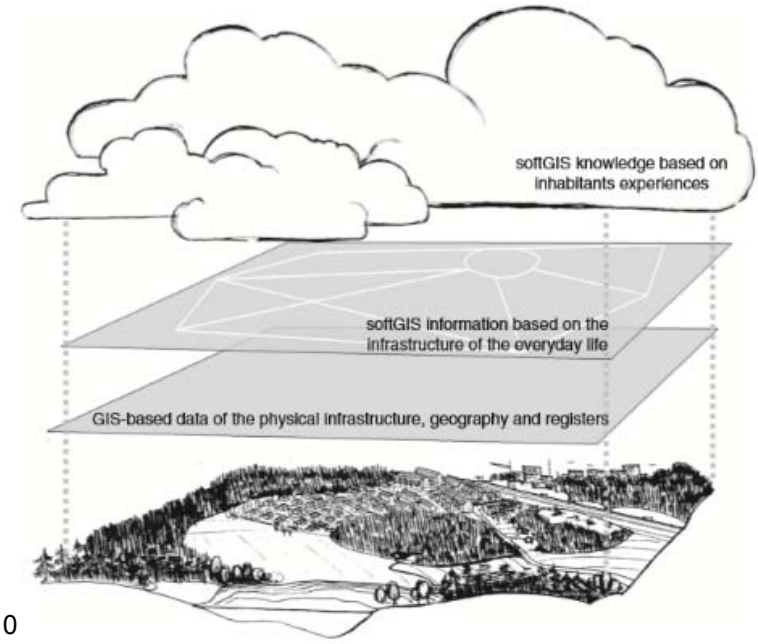
Rebecca K. Bicksler
April 28, 2017

Participatory GIS (PGIS) is sometimes also referred to as "soft" GIS (Kahila and Kyttä, 2010; Ratanen and Kahila, 2009) and "Resident Generated GIS" (Talen, 1999). The main goal of PGIS is to include participatory information in the decision-making process, and "to support the use of local knowledge in research, communicative planning practices and long-span development processes" (Rantanen and Kahila, 2009; Brown and Kyttä, 2014; McCall and Dunn, 2012). PGIS ***combines resident perceptions with spatial information*** so that they can be viewed side by side ***with equal importance*** (Ratanen and Kahila, 2009; Talen 1999).

There are two main driving forces for using a PGIS: community empowerment and innovative data collection. Methods which focus on advanced technology to collect data and display data are referred to as PPGIS; whereas, the focus of PGIS is usually community empowerment, social justice, and inclusion of local knowledge in the discussion of development decisions (Brown and Kyttä, 2014; McCall and Dunn, 2012; Ratanen and Kahila, 2009; Kahila and Kyttä, 2010; Talen 1999; Rambaldi, Kwaku Kyem, McCall, and Weiner, 2006).

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Fig. 19.8 The various layers representing the experienced, lived and physical Environment (*see Color Section 124*)

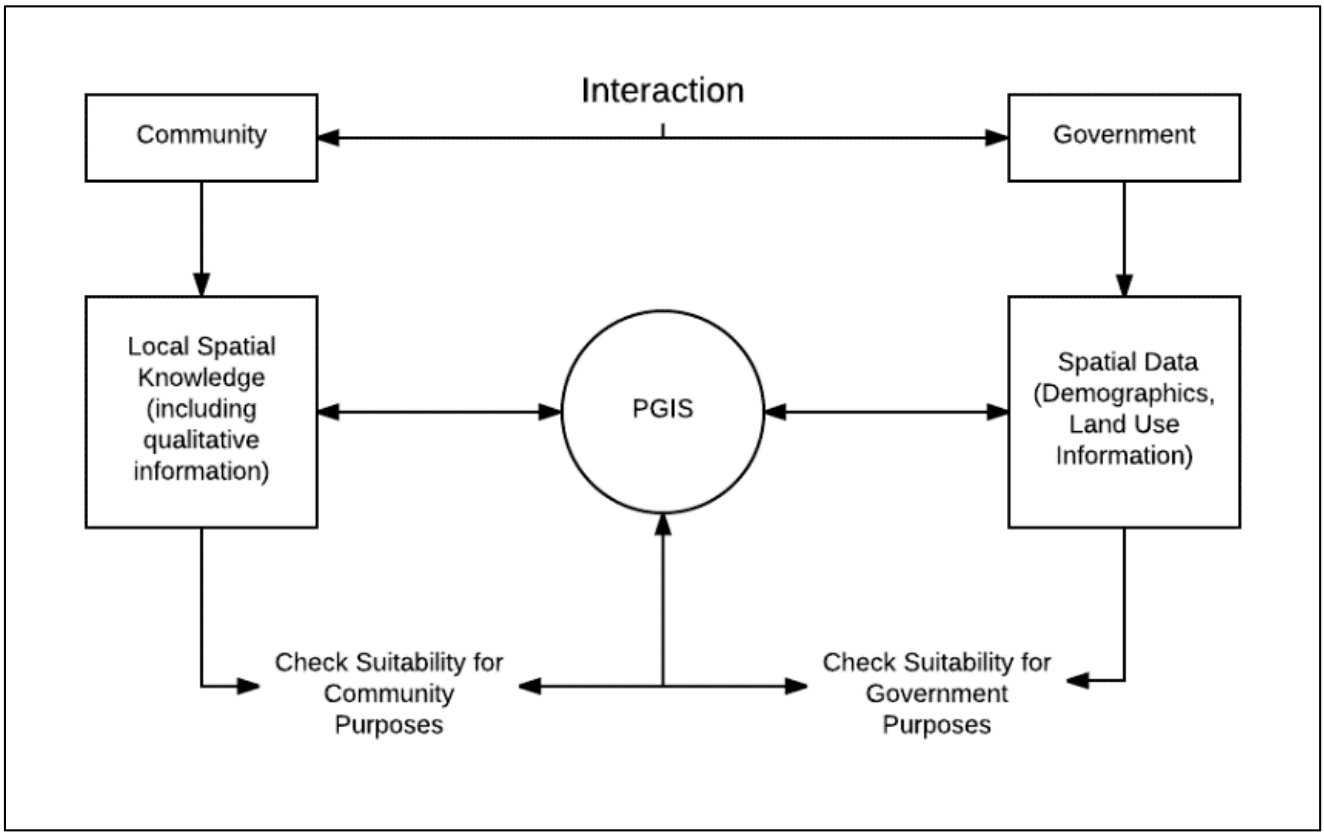


Source: Kahilla and Kyttä, 2010

PGIS is effective because it allows residents to "express their views and perceptions of the neighborhood in a highly visual, dynamic, and spatially contextualized format" (Talen, 1999). In the past, local knowledge have often been left unmapped or unrecorded as it was

viewed as opinion or belief (Ratanen and Kahila, 2009 citing Fisher, 2000 and Puusteinen, 2006), and in some cases local knowledge that is important for the local economy, such as "foods, medicinal herbs, and materials such as dyes or fibers" are left out of official analysis (McCall and Dunn, 2012). However, McCall and Dunn argue that "local spatial knowledge is often more reliable than scientific knowledge 'because it embodies generations of practical essential knowledge, and it operates in interactive, holistic systems'" (McCall and Dunn, 2012 quoting McCall and Mianng, 2005).

PGIS and Community/Government Relationships



Adapted from: https://www.geospatialworld.net/article/should-pgis-serve-two-masters/

The PGIS process is completed through a variety of methods. In many cases, PGIS is completed through a participatory PRA process and transferred to the maps. The layering capabilities of GIS allow multiple voices and perspectives to be viewed simultaneously. Soft GIS methodology creates a connection through the use of "affordances," which are the "perceived opportunities and restrictions concerning a person's actions in a given environment… [It] is not a characteristic of the environment, or a characteristic of the individual, but rather it is something between them." (Kahila and Kyttä, 2010; Ratanen and Kahila, 2009). Similarly, Talen explains that questions such as: 'Where are the roads most congested?' becomes 'What streets do I view as undesirable owing to traffic congestion?" (Talen, 1999). This concept of "affordances" allows researchers and decision makers to consider resident perceptions, or local knowledge, in their

analysis; whereas, typical neighborhood research relies on standard indicators for socioeconomic data (Talen, 1999, citing others).

There are many suggestions in the literature for creating a successful GIS project:

- Quality of data is inextricably linked to sampling design and participation rates (Brown and Kyttä, 2014)
- Careful consideration of inclusion in every stage (McCall and Dunn, 2012)
  - "Participation cuts across the process from gaining a clear understanding of the existing legal and regulatory frameworks, to jointly setting project objectives, defining strategies and choosing appropriate geo-spatial information management tools" (Rambaldi, Kwaku Kyem, McCall, and Weiner, 2006)
  - "Flow of information between different actors is critical" (Kahila and Kyttä, 2010)
- Respect the integrity of local knowledge (McCall and Dunn, 2012)
- Choosing a scale for the information that meets demands for "coverage, comprehensiveness and inclusion on one hand, and on the other hand, information digestion (rather than overload), simplicity and comprehension" (McCall and Dunn, 2012)
- The ability to locate affordances on a map (Brown and Kyttä, 2012; Talen, 1999)
- "Feasibility, local manageability, technical sustainability, and currency, resulting also in cost-effectiveness" (McCall and Dunn, 2012)
- Customization (Brown and Kyttä, 2012)
- Good governance – "accountability, legitimacy, transparency, responsiveness, participation, respect for rights, equity, local usability, and other dimensions" (Rambaldi, Kwaku Kyem, McCall, and Weiner, 2006 citing McCall 2003 and McCall et al. 2005)
- Willingness of decision-makers to use the knowledge and methods in their work (Ratanen and Kahila, 2009, Brown and Kyttä, 2012)

Although PGIS has been used around the world for various projects, the literature and studies remain quite sparse. Most of the examples of PGIS are done by NGOs and community groups, and are not recorded or published. Brown and Kyttä report that "there have only been two longitudinal PGIS studies (Brown & Donovan, 1013b; Brown & Weber, 2011) and one meta-study using PPGIS data (Brown, 2013a)" (Brown and Kyttä, 2012).

PGIS is often seen as an element of decision-support rather than a tool for decision analysis (Greene, Devillers, Luther, and Eddy, 2011) PGIS users can reinforce empowerment through the exploratory function of GIS-based decision tools (McCall, 2003 citing Carver 2001), and these tools will aid in fulfilling the bottom-up planning, democratic empowerment, and advocacy (McCall, 2003 citing Weiner & Harris, 2002 and Gonzalez, 2000; Brown and Kyttä, 2012; Rantanen and Kahila, 2009). Brown and Kyttä state that "the potential of PPGIS as a foundation for iterative public participation, rather than a singular by-product of a public participation process, appears large. Future research could explore the use of PPGIS throughout a complete planning cycle, from scoping to alternative development to decision and monitoring" (Brown and Kyttä, 2012).

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